# CariSurg MedTech Pathways Research Project
# Preliminary Proposal

**Project Title:** Caribbean ED Validation of Machine-Learning Triage Support

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**Statement of the Problem:**

Most AI-based triage tools have been developed and tested outside Caribbean emergency department settings, making local safety, fairness, and effectiveness uncertain. (1–6) At Mercer General Hospital, differences in patient populations, staffing, resources, diagnostic delays, ward-bed availability, clinical judgement, and existing workflows may affect AI-supported triage. (7–9) Therefore, this project addresses whether a machine-learning triage support tool can be locally validated and integrated alongside existing triage practice to improve high-risk patient identification while avoiding increased workload, alert fatigue, under-triage, over-triage, documentation burden, or patient safety risks within a realistic Caribbean ED context. (2,8,10–15)

**Previous Work:**

Tyler et al. and Da'Costa et al. reviewed the use of artificial intelligence and machine learning in emergency department triage. Both studies found that AI has potential to support patient prioritisation, risk prediction, resource allocation, and patient flow management. They also highlighted challenges including algorithmic bias, limited dataset diversity, clinician trust, accountability concerns, and the need for further validation before widespread implementation. Together, these reviews suggest that AI may provide useful decision support, but successful deployment requires careful evaluation within local clinical workflows.(1,2)

Araouchi and Adda, Kang et al., and Bavali-Gazik et al. developed machine-learning models to support emergency department triage and patient prioritisation. Across these studies, models such as stacking approaches, gradient boosting methods, and fusion models achieved strong predictive performance using routinely collected triage data. Despite these promising results, all three studies relied on data from individual institutions or specific patient populations, limiting confidence in how well the models would generalise to other healthcare environments. These findings reinforce the need for local validation before implementation in Caribbean emergency departments.(3,5,6)

Alomari et al. and Wang et al. evaluated the use of large language models for emergency triage and patient guidance. Both studies reported encouraging performance under controlled conditions, with GPT-based systems demonstrating strong agreement with existing triage approaches. However, limitations included reliance on structured scenarios, dataset-specific conditions, and reduced performance when compared with experienced clinicians. These findings suggest that large language models may serve as decision-support tools but should not replace clinical judgement.(4,10)

De Freitas et al. explored patient flow in a Caribbean emergency department using qualitative methods including observation, process mapping, and informal interviews. The study found that patient flow was influenced by work processes, emergency department layout, material resources, staffing levels, non-clinical personnel, and links with other hospital departments. Although the study did not focus on AI, it provides important regional context by demonstrating how local workflow characteristics influence emergency department performance. This highlights why any AI triage support system should be validated within the healthcare environment where it will be used.(7)

Farrohknia et al. conducted a systematic review of emergency department triage scales, including the Emergency Severity Index (ESI), Canadian Triage and Acuity Scale (CTAS), Manchester Triage System (MTS), and Australasian Triage Scale (ATS). The review found that five-level triage systems generally demonstrated better reliability and validity than older three-level systems. However, the authors also noted that evidence supporting some triage components remained limited and varied across healthcare settings. This study is useful because it highlights the importance of understanding the strengths and limitations of existing triage frameworks before introducing AI-supported decision tools.(11)

Christ et al. and Tanabe et al. examined modern emergency department triage systems and the use of the Emergency Severity Index. Their findings showed that structured five-level triage systems can support patient prioritisation and help predict resource utilisation within emergency departments. Tanabe et al. further demonstrated that ESI scores were associated with resource consumption and length of stay, supporting their operational value. Together, these studies provide evidence that existing triage systems offer a useful foundation upon which AI-supported decision tools can be built.(12,13)

Banerjea and Carter investigated waiting and interaction times in a developing-country emergency department in Barbados. They identified significant delays associated with triage, physician assessment, laboratory investigations, and admission processes, demonstrating that patient flow is affected by multiple operational bottlenecks. The study provides Caribbean evidence that emergency department performance depends on more than clinical decision-making alone. This is relevant because any AI triage support system must operate within these existing workflow constraints.(9)

Cho et al. examined emergency department triage decision-making and the role of clinical information in patient prioritisation. The study highlighted that triage decisions are influenced by patient presentation, available information, clinician judgement, and the need for reassessment as patient conditions change. This is relevant to the proposed project because an AI-supported triage tool would need to fit within these existing decision-making processes rather than replace them. It also supports the need to consider workflow integration, staff usability, and patient safety when introducing decision-support tools into emergency department triage.(8)

Abu Arra et al. examined factors influencing nurses' clinical decision-making in emergency departments, while Johansson et al. investigated alarm fatigue among emergency department staff. Together, these studies highlight the influence of workload, organisational factors, information availability, alarm burden, and cognitive demands on clinical performance. The findings suggest that emergency clinicians already operate in information-rich and time-sensitive environments where excessive alerts may reduce effectiveness. These considerations are important because AI-supported triage systems should enhance decision-making without increasing cognitive workload or contributing to alert fatigue.(14,15)

**Gaps Identified:**

The first gap is limited validation of AI triage tools within Caribbean emergency department settings. Several AI triage studies reported strong performance, but many were developed using data from single hospitals, specific countries, structured scenarios, or narrow patient groups. This means Mercer General Hospital cannot assume that a model developed elsewhere would perform safely, fairly, or accurately for Caribbean patients, local staffing patterns, resource availability, and emergency department workflows. Caribbean patient-flow studies also show that local delays, staffing constraints, diagnostic bottlenecks, and ward-bed availability can strongly affect how any new triage support tool would work in practice. (3–7,9)

The second gap is limited evidence on practical workflow integration and human-factor constraints. Many AI triage studies focus mainly on model accuracy, while fewer explain how the tool would fit into existing triage systems such as ESI or how it would affect nurses, doctors, documentation burden, alert fatigue, and patient flow. This is important because triage decisions are shaped by clinical judgement, workload, available information, resource needs, and reassessment during waiting. For Mercer General Hospital, the proposed tool should therefore be tested as a support layer beside existing triage methods, with attention to high-risk patient identification, under-triage, over-triage, staff usability, cognitive workload, and workflow fit.(1,2,8,10–15)

**Proposed Solution:**

To address the limited validation of AI triage tools in Caribbean emergency department settings and the limited evidence on practical workflow integration, this project proposes a 12-week pilot study to test a machine-learning triage support tool using de-identified emergency department data from Mercer General Hospital. The tool would not replace nurses, doctors, or existing triage systems. Instead, it would act as a clinically supervised support layer beside current triage practice, helping staff identify patients who may be at higher risk while keeping the process familiar, practical, and guided by clinical judgement. (1,2,7–9)

The pilot would use routinely collected triage information such as age, sex, vital signs, chief complaint, mode of arrival if available, and existing triage category. A simple baseline model, such as logistic regression or a decision tree, would be developed first because these models are easier to interpret, explain, and compare with current triage decisions. If time allows, the baseline model could then be compared with stronger machine-learning models such as Random Forest or Gradient Boosting to assess whether more complex models improve performance enough to justify their use in a clinical support setting. (3,5,6)

The model would be evaluated using clinically relevant measures rather than accuracy alone. These would include high-acuity recall, under-triage risk, over-triage rate, agreement with existing triage outcomes, and the model's ability to identify patients who may need urgent review, admission, or higher-level care. This is important because an AI triage tool must prioritise patient safety by reducing the risk of missing high-risk patients, while also avoiding unnecessary pressure on already limited emergency department resources. (1,2,4,10)

To address the workflow integration and human-factor gap, a simple workflow mock-up would also be created to show how the tool could fit into Mercer General Hospital's existing triage process. The mock-up would show where a risk score or alert could appear, how it could sit beside an existing triage framework such as ESI, and how nurses and doctors might interpret the output without adding unnecessary documentation work. The design would also consider clinical reassessment, alert fatigue, cognitive workload, staff usability, and patient flow during waiting periods. This is important because Caribbean emergency department performance is influenced by local staffing, diagnostic delays, physical layout, material resources, ward-bed availability, and links with other hospital departments. (7–9,11–15)

The pilot would remain within a safe 12-week scope. It would use only de-identified data and would not involve live hospital deployment, autonomous decision-making, regulatory approval, or direct changes to patient care. Any future clinical use would require stronger validation, clinical oversight, stakeholder feedback, and hospital or ethics approval.

## Workflow and AI Plug-in Points

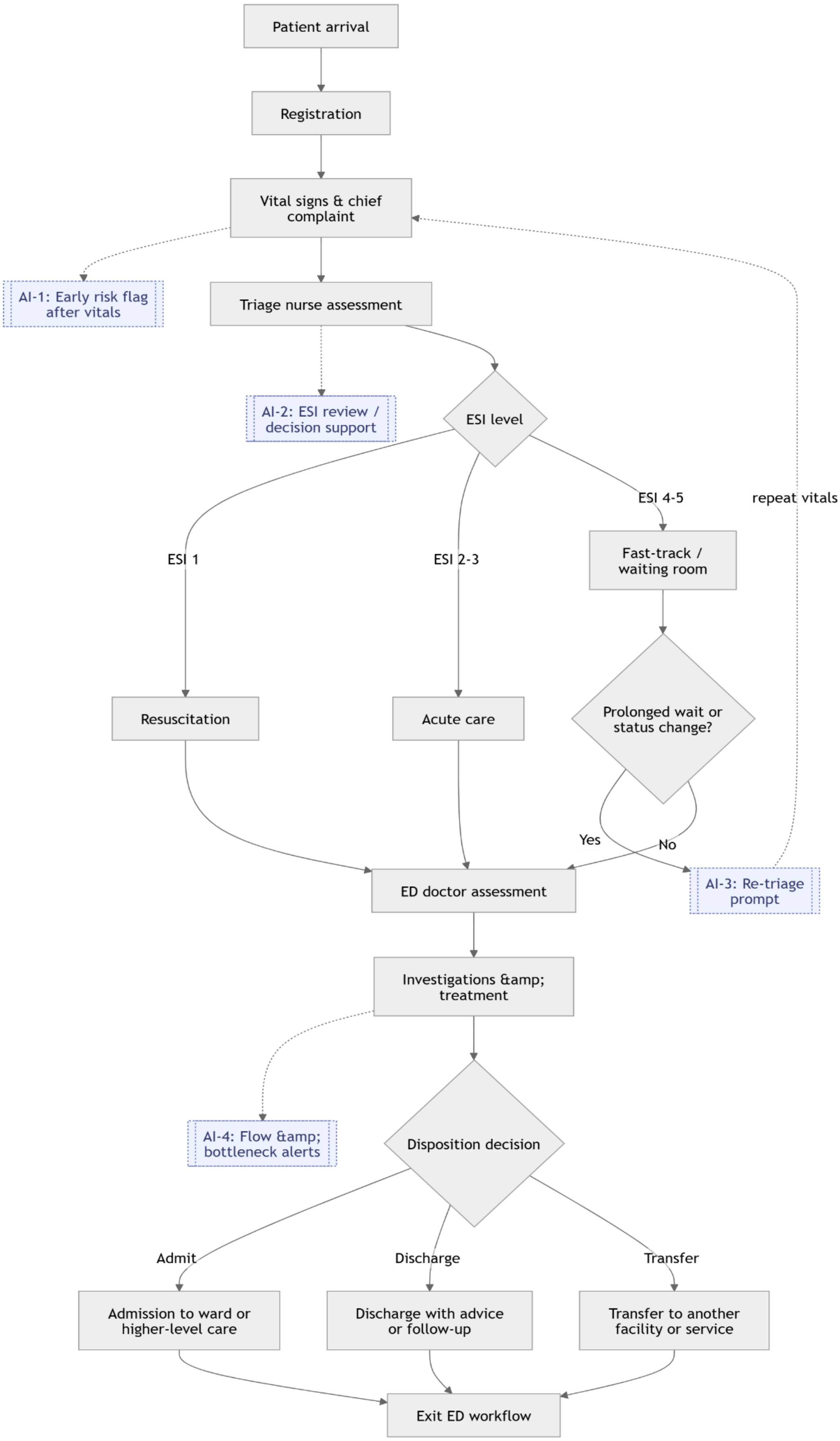


*Figure 1:Proposed workflow position of a machine-learning triage support tool within the Mercer General Hospital emergency department triage process.*

The proposed system would fit within the existing Mercer General Hospital emergency department workflow rather than replacing the current triage process. As shown in Figure 1, the workflow begins with patient arrival by ambulance, walk-in, private vehicle, taxi, or EMS. This is followed by registration, vital-sign collection, triage nurse assessment, Emergency Severity Index (ESI) assignment, care-area allocation, ED doctor assessment, investigations or treatment, disposition, and exit from the emergency department.

The first AI plug-in point occurs after vital-sign collection. At this stage, the system could use information already available during triage, such as age, sex, arrival mode, chief complaint, vital signs, pain score, allergies, and brief clinical history. The output would be an early risk flag for abnormal vital-sign patterns or possible high-risk presentation. This would support the triage nurse's assessment while keeping the final decision under clinical control.

The second AI plug-in point occurs after the ESI level is assigned. At this stage, the system could compare the assigned ESI category with the model's estimated risk level to identify possible mismatch. For example, it could flag patients who may have been under-triaged or over-triaged based on their recorded observations and presenting features. This would allow the nurse or doctor to review the case without replacing the existing ESI framework.

The third AI plug-in point occurs during waiting-room or fast-track monitoring. Since triage is not a one-time event, patients may deteriorate while waiting, especially during periods of high patient volume or delayed assessment. The system could monitor waiting time, initial ESI level, and previously recorded observations to prompt repeat vital signs or reassessment when appropriate. This would support re-triage and help reduce the risk of missing patients whose condition changes after initial triage.

The fourth AI plug-in point occurs during investigations, treatment, and patient-flow monitoring. At this stage, the system could highlight delays linked to laboratory investigations, imaging, specialist consultation, or ward-bed availability. This function would not make clinical decisions independently but could support charge nurses and doctors by improving visibility of bottlenecks that affect patient movement through the department.

Overall, the workflow diagram positions the proposed AI system as a low-friction decision-support layer within existing clinical practice. Its role would be to support early risk recognition, ESI review, waiting-room reassessment, and patient-flow awareness while maintaining clinician oversight, avoiding unnecessary documentation burden, and respecting the operational realities of a Caribbean emergency department.

**GitHub Repository:** https://github.com/Kari-2005/carisurg-portfolio

The Week 3 workflow diagram source file and exported diagram are available in the repository under docs/Week3/.

**Constraints & Stakeholders:**

Three main workflow constraints must be considered before introducing an AI-supported triage tool at Mercer General Hospital.

The first constraint is paper-first documentation and delayed digital data availability. Patient information and vital signs may initially be recorded on paper before being entered into an electronic system. This creates a delay between when information is available to clinicians and when it becomes available digitally. An AI system that relies entirely on real-time electronic health record data may therefore be unreliable at the point where triage decisions are being made. To reduce this risk, the proposed tool should use information already collected during triage and support gradual structured digital data capture.

The second constraint is limited triage nurse time and cognitive workload. Triage nurses work under significant time pressure, especially during periods of high patient volume. Additional data entry, complex interfaces, or excessive alerts could increase workload and disrupt existing triage practice. If the tool slows staff down or produces too many alerts, clinicians may ignore it or avoid using it. Therefore, the system should function as low-friction decision support, rely mainly on existing triage information, and present concise, clinically relevant outputs.

The third constraint is patient deterioration during waiting and the need for re-triage. Patient conditions may change after the initial triage decision, particularly during prolonged waits. A model that only supports the first ESI assignment may miss deterioration later in the patient journey. The proposed system should therefore support repeat observations and generate prompts for reassessment when appropriate, while keeping final decisions under clinical supervision.

*Table 1:Workflow constraints, design impact, and mitigation strategies for the proposed AI-supported triage workflow.*

| Workflow constraint | Design impact | Mitigation |
|---|---|---|
| Paper-first documentation and delayed digital data availability | The model may not access triage information at the exact point when decisions are being made. | Use information already collected during triage and support gradual structured digital data capture. |
| Limited triage nurse time and cognitive workload | Extra data entry, complex screens, or excessive alerts may cause staff to ignore or avoid the tool. | Keep the tool low-friction, concise, and based mainly on existing triage information. |
| Patient deterioration during waiting and need for re-triage | A model based only on initial triage may miss patients whose condition worsens while waiting. | Include prompts for repeat observations and reassessment when appropriate, with final decisions kept under clinical supervision. |

*Table 2:Stakeholder map for integrating a machine-learning triage support tool into Mercer General Hospital's emergency department workflow.*

| Stakeholder | Main concern | What success looks like |
|---|---|---|
| Triage nurse | The tool must not slow triage, increase documentation burden, or undermine clinical judgement. | The nurse receives a clear risk flag using information already collected during triage and can make the final decision quickly. |
| ED doctor / consultant | The tool must provide reliable and clinically defensible risk information. | High-risk patients are easier to identify for urgent review without replacing clinical assessment. |
| Nurse-in-charge / charge nurse | The tool must improve visibility of patient flow without adding confusion during busy shifts. | Waiting-room risk, bottlenecks, and staff workload are easier to monitor and act on. |
| Registration / administrative staff | The tool must not delay intake or create duplicate data-entry work. | Patient information is captured accurately once and can support triage without extra administrative burden. |
| Patient advocate / patient representative | The tool must protect patient safety, fairness, privacy, and timely care. | Patients are prioritised fairly, personal data are protected, and high-risk patients are not missed. |

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